Sampling-Based Hierarchical Trajectory Planning for Formation Flight

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***Abstract*— Formation flight of uncrewed aerial vehicles (UAVs) poses significant challenges in terms of safety and formation keeping, particularly in cluttered environments. However, existing methods often struggle to simultaneously satisfy these two critical requirements. To address this issue, this paper proposes a sampling-based trajectory planning method with a hierarchical structure for formation flight in dense obstacle environments. To ensure reliable sharing of local sensing information among UAVs, each UAV generates a safe flight corridor (SFC), which is transmitted to the leader UAV. Subsequently, a sampling-based formation guidance path generation method is designed as the front-end strategy, steering the formation to fly in the desired shape safely with the formation connectivity provided by the SFCs. Furthermore, a model predictive path integral (MPPI) based distributed trajectory optimization method is developed as the back-end part, which ensures the smoothness, safety and dynamics feasibility of the executable trajectory. To validate the efficiency of the developed algorithm, comprehensive simulation comparisons are conducted. The supplementary simulation video can be seen at https://www.youtube.com/watch?v=W2hXrjhl864**

***Index Terms*— Multi-UAVs, formation keeping, trajectory planning, obstacles avoidance.**

I. Introduction

THE utilization of multi-UAV systems has gained significant traction in both industrial and research domains, owing to their inherent robustness and efficiency, surpassing that of individual UAVs. Collaborative efforts among UAVs have successfully accomplished numerous tasks, including swarm surveillance [1], rescue missions [2], goods delivery [3], tracking [4], environmental exploration [5]. In the aforementioned applications, it is imperative for the multi-UAV system to form a desired configuration and maintain coordinated movement within that formation. When navigating through the clustered environment, the UAV swarm faces three primary challenges. First, all UAVs must avoid collision with obstacles, which is a fundamental safety requirement for formation flight. Additionally, there is a risk of internal collision within the swarm if the inter-UAV distances become too small; thus, reciprocal collision avoidance must be addressed to ensure safe formation flight. Second, the formation shape determined by the positions of the UAV swarm may become distorted during flight, requiring real-time adjustment based on the continuously updated positions of all UAVs. Third, conflicts can emerge between safety constraints and the requirement to preserve formation shape, especially in obstacle-dense environments. Addressing these competing demands is critical for effective and safe UAV swarm navigation in unknown environments.

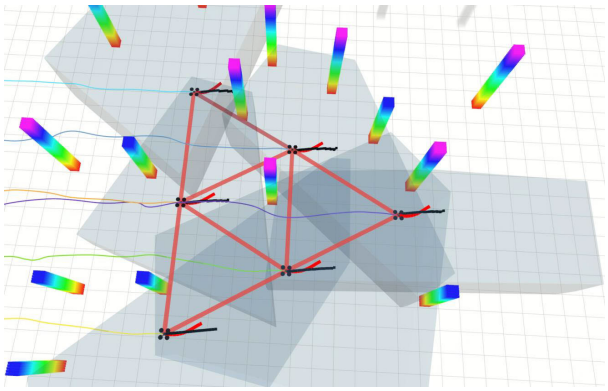


Fig. 1. A snapshot during the process of formation flight. The light grey polytope is the generated SFC by each UAV, the black lines and light red lines are the formation guidance path generated in Section IV and the trajectory optimized in Section V respectively.

To address the above challenges, a hierarchical formation trajectory planning method is proposed in this paper. A snapshot of the formation flight process can be seen in Fig. 1, where the proposed algorithm is applied. The framework of the proposed method is illustrated in Fig. 2. Each UAV independently executes the state estimation, local mapping and SFC generation modules. The essential state information, including local position, euclidean distance transform (EDT) and SFC is generated with the local data from sensors. Utilizing the SFC, local position and the minimum distance to the nearest obstacles shared by all the UAVs in the communication network, the formation guidance paths are derived through the centralized formation guidance path planning module running on the leader, which are then broadcast to each UAV from the leader. The union of SFCs from all UAVs is utilized to restrict the whole formation safe region, thereby adapting the clustered environments. Benefiting from the formation connectivity provided by the formation safe region, the safe formation guidance paths ensure the maintenance of formation

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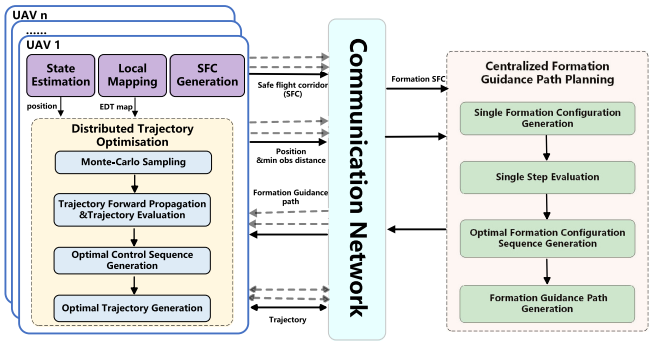


Fig. 2. Framework of the sampling-based hierarchical trajectory planning method.

during the flight process without any collisions with obstacles. Finally, the feasible trajectory is generated locally based on the guidance path, using the distributed trajectory optimization module for each UAV.

The key contributions of this paper can be summarized as follows:

1) A centralized formation guidance path generation method is proposed as the front-end path planning algorithm based on the sampling and receding horizon idea, ensuring flight safety while the formation maintains the desired shape in clustered environments.
2) A distributed trajectory optimization strategy is developed as the back-end trajectory optimization algorithm based on MPPI method, which guarantees the feasibility of the trajectory while following the formation guidance path.

The structure of this article is outlined as follows. The review of related works is presented in Section II. Section III offers a comprehensive introduction to fundamental concepts and principles that serve as foundations for comprehending the subsequent algorithm. The sampling-based formation guidance path generation and trajectory optimization components are elaborated in Section IV and Section V, respectively. The performance of the method is validated through simulations, as presented in Section VI. Finally, the article concludes with a summary and concluding remarks in Section VII.

II. Related Works

The majority of existing methods for formation keeping in multi-UAV systems employ control techniques such as position-based control [6], displacement-based control [7], and distance-based control [8]. However, these methods typically result in rigid formations that lack scalability and are not adaptable to obstacle-rich environments. To address this limitation, a complex Laplacian-based control method with a leader-follower framework has been proposed [9]. This approach utilizes graph theory [10] and leverages the complex Laplacian theorem, where accurate formation information is only available to the selected leader, while the other UAVs receive shape information only. For achieving steady-state collective motion, the weight-modification mechanism has been introduced to the complex Laplacian-based formation control [11]. Additionally, bearing-based formation control has gained attention due to advancements in sensor technology, allowing for more precise bearing information [12]. However, it is important to note that bearing-based control methods do not address orientation tracking. To handle more general transformations such as translation, scaling, shearing, or their compositions, the affine formation control approach based on the stress matrix, a type of more general Laplacian matrix, has been proposed [13]. Subsequently, the concept of modularity has been introduced in [14], eliminating the need for a designated leader. In this approach, each module defines a cost based on least-squares affine alignment between the positions of the robots in the current and nominal configurations. Although these methods effectively achieve the desired formations with good performance, collision avoidance is not explicitly considered. In [15], a distributed control strategy has been devised to autonomously achieve a desired 3D formation in a team of agents, while also accounting for collision avoidance.

In addition to traditional control methods, trajectory planning plays a significant role in addressing the formation problem. One approach based on trajectory planning is presented in [16], where a shape-vector is introduced to facilitate the maintenance of the overall formation planning. To enhance practicality, the state estimation module is extended with visual-inertial capabilities, enabling formation execution without relying on an external motion capture system or GPS in [17]. Besides, in [18] the formation trajectories are generated by using the admittance model. Furthermore, a replanning mechanism is introduced to the formation trajectory planning in [19], which can reconstruct the formation.

The aforementioned methods perform impressively in obstacle-free environments. However, clashes between formation maintenance and obstacle avoidance can occur in clustered environments, which cannot be addressed with those methods. To tackle this issue, some techniques attempt to co-optimizing formation maintenance with other aspects of trajectory performance. For instance, a virtual-structure based approach for controlling a group of quadrotors carrying out agile interleaved maneuvers is proposed in [20]. A list of potential fields are constructed, including formation preservation, obstacle aversion and collision avoidance fields. The resultant optimal velocity vector is achieved through the sophisticated optimization of these potential fields. Nevertheless, local minima trapping is a noted drawback of the potential field method. In contrast, in [21], model predictive control (MPC) is employed for the concurrent resolution of optimal assignments and collision-free trajectories. Real-time computations enable instantaneous formation adaptations, allowing for prompt responses to communication or tracking discrepancies. Taking another route, a differentiable Laplacian-based metric is introduced in [22], facilitating the quantification of formation keeping performance. However, this approach relies heavily on the weight parameters to coordinate the overlapping effects of formation, obstacle aversion, and internal collisions.

To avoid the unsatisfactory comprise of formation in the co-optimization, various methods adopt a hierarchical framework, which is also applied in our work. In [23], a valid formation configuration space using an undirected graph is firstly

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determined, and a graph search methodology is adopted to ascertain a feasible path for the entire system. However, this approach is inherently constrained to environments equipped with a prior map. In contrast, in [24] a global-local controller is implemented to ensure obstacle-free formation flight. The global controller is responsible for generating the guiding virtual structure, which serves as a safe region for navigation. Concurrently, the local controller ensures adaptive formation within the dynamic structure that is engendered by the global controller. A similar idea is identifiable in [25], wherein the formation configuration is primarily determined within the formation polytope. Afterwards, each UAV autonomously navigates toward the designated optimal target independently. This idea is further expanded in [26] and [27], where the distributive computation is applied to both UAV formation flight and object transportation tasks. In dense obstacle environments, the safe region generated may be overly constrained, which could impede the accommodation of the entire formation at a reduced scale. This limitation could potentially dismantle the formation integrity. To counteract this issue, the "split and merge" concept is introduced to enhance formation coordination in [28]. In this way, the entire formation has the capacity to decompose into smaller sub-formations corresponding to the emergent child regions. Conversely, when the safe region can accommodate the big formation, these sub-formations can seamlessly reassemble into a cohesive full formation. To expand the safe formation region, the proposed work constructs the restricted area using the union of SFCs, rather than the intersection approach employed in [25], [26], [27], and [28].

Although substantial research has been conducted on UAV formation, majority of these studies primarily focus on either formation control [9], [10], [11], [12], [13], [14] or obstacle and collision avoidance [15]. Besides, the integrated approaches of formation coordination and safety in methods [20], [21], [22], [23], [24], [25], [26], [27], [28] cannot guarantee effective formation keeping, especially when the available safe region around the formation is excessively restricted.

III. Preliminaries

A. Formation Definition

In this part, some essential definitions are given for subsequent discussions. Considering a formation $\mathcal{F}$ consisting of N UAVs ($N \in \mathbb{Z}^+$). The desired formation shape, denoted as $\mathcal{F}_{des}$, is predetermined and represented by a set of N desired individual positions $\boldsymbol{p_d^i} = \left[x_d^i, y_d^i, z_d^i\right] \in \mathbb{R}^3$. These positions are defined relative to the formation center $\boldsymbol{c} = \left[c_x, c_y, c_z\right] \in \mathbb{R}^3$. Throughout the flight, the formation center is subject to change as the UAVs progress towards their respective goals, and adjustments in the formation scale may occur. Therefore, the formation can also be characterized by the formation configuration $\boldsymbol{f_c}$, incorporating a scale factor $s \in \mathbb{R}^+$, the set of desired relative positions $\boldsymbol{P_{des}} = \left\{\boldsymbol{p_d^1}, \ldots, \boldsymbol{p_d^N}\right\}$, and the formation center $\boldsymbol{c}$. The desired relative position set remains constant throughout the entire process, and for simplicity, we exclude the variable $\mathbf{P_{des}}$ from the formation configuration $\boldsymbol{f_c}$. Instead, we summarize formations with the same $\boldsymbol{P_{des}}$ as similar formations. Therefore, the formation configuration can be represented as $\boldsymbol{f_c} = \{s, \boldsymbol{P_{des}}, \boldsymbol{c}\}$. Furthermore, the single formation position $\boldsymbol{p_f^i}$ for UAVi within a formation can be calculated as follows:

$$\boldsymbol{p_f^i} = \boldsymbol{c} + s\boldsymbol{p_i^d}, i \in \{1, \cdots, N\} \tag{1}$$

B. Simplified Description of Obstacle-Free Space

The whole 3-D workspace is given by $\mathcal{W} \subseteq \mathbb{R}^3$,and the obstacle and unknown areas are given by $\mathcal{O}_{obs}$ and $\mathcal{O}_{unk}$ respectively. Therefore, the collision-free workspace is given by $\mathcal{W}_{free} = \mathcal{W} \setminus \mathcal{O}_{obs} \setminus \mathcal{O}_{unk}$. Each UAV $i \in \{1, \cdots, N\}$ in $\mathcal{F}$ can sense its local environment $\mathcal{W}_i$ independently and the local collision-free workspace is denoted as $\mathcal{W}_{free_i}$.

The surrounding environment can be recorded with many existing methods of map constructing such as truncated signed distance field map [29], euclidean signed distance field map [30] and EDT map [31]. However, due to the substantial volume of data contained within the above representations of map, it is impractical to transmit this information in real-world between UAVs. The construction of the 3-D SFC serves as an alternative method to map the environment with limited data volume, which is a simplified representation of the free space $\mathcal{W}_{free_i}$ surrounding each UAVi. The SFC, also known as the safe convex polytope $\mathcal{P}_i$ of UAVi, represents the region within which safe flight is ensured. The convex polytope $\mathcal{P}_i$ can be represented as follows:

$$\mathcal{P}_i = \left\{\boldsymbol{p} \in \mathbb{R}^3 \mid A\boldsymbol{p} \leq \boldsymbol{b}, \text{ for } \boldsymbol{p} \in \mathbb{R}^{n_l \times 3}, \boldsymbol{b} \in \mathbb{R}^{n_l}\right\} \tag{2}$$

where n_l denotes the number of faces of $\mathcal{P}_i$, A and b are the description parameters of a sequence of half spaces and $\boldsymbol{p}$ is the position of point inner $\mathcal{P}_i$. The only information needed to be shared is the polytope description parameters, whose volume is significantly smaller compared to a local map.

By using the current position of each UAV and the local environmental information from sensors as inputs, the local safe flight corridor is computed through the fast inflation and recursive intersection algorithm presented in [32]. This algorithm involves two main stages: ellipsoid inflation and the calculation of the Maximum Volume Inscribed Ellipsoid (MVIE) within the convex polytope. The ellipsoid inflation process is depicted in Fig. 3. Initially, the ellipsoid is expanded from a given starting ellipsoid ε_0 (or MVIE in the previous iteration). As the ellipsoid ε inflates, it performs collision checks against each obstacle $\mathcal{O}_{obs_i} \in \mathcal{O}_{obs}$. Upon intersecting with an obstacle $\mathcal{O}_{obs_i}$, a half space $H = \left\{\boldsymbol{p} \mid \mathbf{a}_0^T \boldsymbol{p} < \mathbf{b_0}\right\}$, where $\mathbf{a}_0^T$ and $\mathbf{b_0}$ are the description parameters in one half space, is generated at the point of intersection. Obstacles located outside of H are marked as checked. This inflation process continues until all sensed obstacles are processed, resulting in a sequence of half-spaces $\mathcal{H} = \{H_0, H_1, \cdots H_n\}$, whose intersection forms the safe convex polytope $\mathcal{P}_i$. Following this, the MVIE is calculated using the affine scaling algorithm and its volume is set as a lower bound for subsequent calculation. To maximize the surrounding free space, $\mathcal{P}_i$ is calculated recursively until the volume of the resulting MVIE no longer exceeds a specified threshold.

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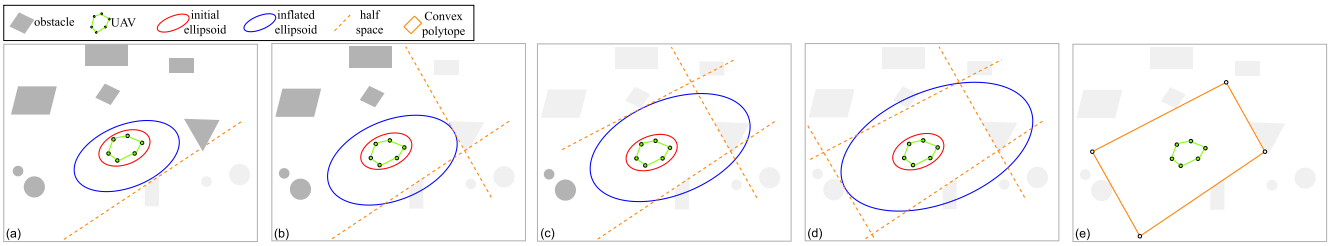


Fig. 3. Illustration of the ellipsoid inflation process. The UAV (green polytope) generates its safe convex polytope by inflating the initial ellipsoid (red ellipsoid). During this process, multiple half-spaces (orange dotted lines) are created. Fig. 3a- Fig. 3d depict the detailed half-space generation steps, and Fig. 3e shows the resulting safe convex polytope (orange polytope).

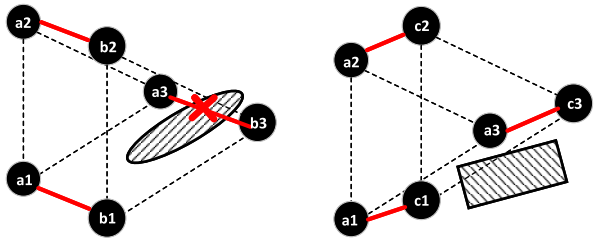


Fig. 4. The black dot lines connect the inner points in a formation configuration and the red lines are the connecting paths. In Fig. 4a, there is no formation connectivity between $\mathbf{f_c^a}$ and $\mathbf{f_c^b}$ due to the intersection of the path between $a3$ and $b3$ with the ellipse obstacle. Conversely, as shown in Fig. 4b, all paths connecting the correlation points between $\mathbf{f_c^a}$ and $\mathbf{f_c^c}$ are unobstructed, thus indicating that the two formation configurations are connected.

C. Formation Connectivity

Inspired by the formation configuration connection outlined in definition 2 of [23], we formulate the formation connectivity based on it, which is illustrated in Fig. 4. Thanks to the safe region confined by the SFC, the formation connectivity can be guaranteed as follows: two formation configurations f_c^m and f_c^n are characterized as connected when they meet the following criteria:

$$\begin{aligned} \boldsymbol{p}^{\boldsymbol{i}}_{f_c^m} &\in \mathcal{P}_i \\ \boldsymbol{p}^{\boldsymbol{i}}_{f_c^n} &\in \mathcal{P}_i \end{aligned} \tag{3}$$

where $\boldsymbol{p}^{\boldsymbol{i}}_{f_c^m}, i \in \{1, \cdots, N\}$ represents the formation target for UAVi in f_c^m. This constraint establishes the connectivity within the scope of the same local $\mathcal{P}_i$ for all UAVs, ensuring a safe straight path for the complete formation to transition from f_c^m to f_c^n.

D. Trajectory Optimization Problem Formulation

Consider the case where all UAVs adhere to a linear dynamic model in differential flat outputs $[p_x, p_y, p_z, \psi]$, where p_x, p_y, p_z denote the position in three-dimensional coordinates and ψ represents the yaw. The dynamic model for each UAV i can be written as:

$$\begin{aligned} &\dot{\boldsymbol{x}}_{\boldsymbol{t}}^{\boldsymbol{i}} = \boldsymbol{A}\boldsymbol{x}_t^i + \boldsymbol{B}\boldsymbol{\mu}_{\boldsymbol{t}}^{\boldsymbol{i}} \\ &\boldsymbol{A} = \begin{bmatrix} \mathbf{0}_{6\times3} & \boldsymbol{a} \\ \mathbf{0}_{4\times3} & \mathbf{0}_{4\times7} \end{bmatrix}, \boldsymbol{B} = \begin{bmatrix} \mathbf{0}_{6\times4} \\ \mathbf{I}_{\mathbf{6}\times\mathbf{6}} \end{bmatrix}, \boldsymbol{a} = \begin{bmatrix} \mathbf{I}_{\mathbf{6}\times\mathbf{6}} \\ \mathbf{0}_{6\times1} \end{bmatrix}^{\mathrm{T}} \end{aligned} \tag{4}$$

where $\boldsymbol{x}_t^i = \left[p_x^i, p_y^i, p_z^i, v_x^i, v_y^i, v_z^i, a_x^i, a_y^i, a_z^i, \psi^i\right]^{\mathrm{T}}$ is the state vector at time t including position, velocity, acceleration and yaw. Here we let $\boldsymbol{\mu}_{\boldsymbol{t}}^{\boldsymbol{i}} = \boldsymbol{u}_t^i + \boldsymbol{v}_t^i$, where $\boldsymbol{u}_t^i = \left[\dot{a}_x, \dot{a}_y, \dot{a}_z, \dot{\psi}\right]^{\mathrm{T}}$ is the control input and $\boldsymbol{v}_{\boldsymbol{t}}^{\boldsymbol{i}} \sim \mathcal{N}(0, \Sigma)$ is the stochastic control input following the Gaussian distribution with 0 - mean value and Σ - variance. During the stochastic diffusion process, $\boldsymbol{v}_{\boldsymbol{t}}^{\boldsymbol{i}}$ is applied for generating the forward sampling.

By utilizing the framework of the path integral control [33], we can sample a sequence of optimal control input $U_i = \left\{\boldsymbol{u}_t^i, \ldots, \boldsymbol{u}_{t+T}^i\right\}$ in a time horizon T to generate a predicted trajectory. In the evaluation of a sampled trajectory, several objectives are considered, including the terminal cost $\phi(\boldsymbol{x}_{t+T}, t+T)$, the running cost $\rho(\boldsymbol{x}_t, t)$, and the control input cost $\chi(\boldsymbol{u}_t)$. To acquire the optimal trajectory, the above costs are minimized by solving U_i. The total cost function of the control problem is defined as follows:

$$\begin{aligned} V(\boldsymbol{x}_t^i, t) = \mathbb{E}_{\mathbb{Q}}[\phi(\boldsymbol{x}_{t+T}^i, t+T) \\ + \int_t^{t+T} (\rho(\boldsymbol{x}_\tau^i, \tau)) + \chi(\boldsymbol{u}_\tau^i) d\tau] \end{aligned} \tag{5}$$

where $\chi(\boldsymbol{u}_\tau^i) = \frac{1}{2}(\boldsymbol{u}_\tau^i)^T \boldsymbol{R}\boldsymbol{u}_\tau^i$, $\boldsymbol{R}$ is a positive definite matrix and $\mathbb{E}_{\mathbb{Q}}$ is the mathematical expectation taken by Monte-Carlo sampling. As a result, the trajectory planning problem can be transformed into a stochastic optimal control problem subject to dynamic model:

$$\begin{aligned} \arg\min_{U_i^*} \quad & V(\boldsymbol{x}_t^i, t) \\ \text{s.t.} \quad & \dot{\boldsymbol{x}}_t^i = A\boldsymbol{x}_t^i + B\boldsymbol{\mu}_t^i \end{aligned} \tag{6}$$

IV. Sampling-Based Formation Guidance Path Generation Strategy

In this section, the sampling-based methodology for generating formation guidance paths is detailed. Firstly, the sampling process used for generating formation configurations is illustrated. Secondly, a comprehensive set of cost functions is introduced, which are utilized to evaluate the obtained formation configurations and formation configuration sequences. Subsequently, the task assignment optimization is performed to optimize the assignment result which is excluded in the formation configuration sequence. Finally, the formation configuration sequence undergoes a transformation into formation guidance paths using path links.

A. Formation Configuration Sample Process

The sampling process is subdivided into two steps to accommodate the dual values inherent in a formation configuration. Initially, a significant quantity of formation center $\boldsymbol{c}$ within

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a hemispherical surface, centered around the local initial formation center c_0 are sampled. This approach is implemented through a step propagation mechanism, with a radius specified by $r = (1+\gamma)r_{\min}$. where $r_{\min}$ denotes the predefined minimal propagation radius, and $\gamma \in [0,1]$. The propagation is strategically directed towards the target centroid $\boldsymbol{c_{goal}}$ in the target formation configuration $\boldsymbol{f_c^{goal}}$. Subsequently, the scale set S is obtained by uniformly generating scale s values within a limited range encapsulated by the minimum and maximum scale thresholds $s_{\min}$ and $s_{\max}$. Each of member in S is then paired with a corresponding $\boldsymbol{c}$ to form a plethora of sampled formation configurations. In this way, the whole safe region is discretized into thousands of formation configurations.

B. Formation Configuration Evaluation Functions

To discern the optimality of the sampled formation configurations, a set of cost functions has been developed, thereby achieving an integrated optimum that takes into account the constraints of formation keeping and safety.

1) Goal Driven Cost: The movement of the formation is driven by the setting goal formation configuration. Considering the distance between the current and goal formation center, the goal driven cost is designed as follows:

$$c_g = k_g \left|\left| \boldsymbol{c_{cur}} - \boldsymbol{c_{goal}} \right|\right| \tag{7}$$

where k_g is the weight parameter, $\boldsymbol{c_{cur}}$ is the formation center in the current formation configuration $\boldsymbol{f_c^{cur}}$.

2) Scale Prefer Cost: In addition to specifying the desired shape, the desired formation scale can also be predefined. This supplementary parameter permits differentiation between two formation configurations that display identical levels of safety with the same $\boldsymbol{c}$. Hence, the formation scale should lie within close approximation of the preferred scale:

$$c_s = k_s \left| s_{cur} - s_{des} \right| \tag{8}$$

where k_s is the weight parameter, s_{cur} is the scale in f_c^{cur} and s_{des} is the desired scale.

3) Safe Region Cost: Once all formation configurations in one sequence are connected, the generated guidance paths based on the sequence enable the formation to maintain its desired shape without being disrupted by obstacles. To achieve this, the formation connectivity criteria (3) can be transformed into the safe region cost. For each formation configuration, the safe region cost c_{safe} can be defined as follows:

$$\begin{aligned} c_{\text{safe}} &= \sum_i^N g_{\text{safe}}^i \\ g_{\text{safe}}^i &= \begin{cases} k_{\text{safe}}, & \boldsymbol{p}_f^i \notin \mathcal{P}_i \\ 0, & \boldsymbol{p}_f^i \in \mathcal{P}_i \end{cases} \end{aligned} \tag{9}$$

where k_{safe} is a large punishment cost added to the unsafe formation configuration target, and g_{safe}^i is the self safe cost in terms of each formation target in a formation configuration.

4) Safe Risk Cost: During the traversal of a tightly packed environment, each UAV owns different size of free space to bypass obstacles. To render this metric more concrete, it is repurposed into a measure of risk magnitude, which can be assessed by calculating the minimal distance to the obstacles. In the context of formation maintenance, UAVs that exhibit high levels of risk are characterized by limited mobility due to the constraints imposed by their risk profiles. In contrast, UAVs with lower risk profiles are required to adjust their positions to accommodate those with higher risk levels. To quantify the risk impact, the safety risk cost, denoted as c_{risk}, is designed as follows:

$$\begin{aligned} c_{\text{risk}} &= \sum_i^N g_{\text{risk}}^i \\ g_{\text{risk}}^i &= w_{risk}^i \left|\left| \boldsymbol{f_t^i} - \boldsymbol{p_i} \right|\right| \\ w_{risk}^i &= \begin{cases} \dfrac{d_{obs}^i}{\sum_{j=0}^N d_{obs}^j}, & d_{obs}^i > d_{risk} \\ 0, & d_{obs}^i \le d_{risk} \end{cases} \end{aligned} \tag{10}$$

where g_{risk}^i represents the individual safe cost as appraised for each UAV, $\boldsymbol{f_t^i}$ is the temporary calculated target of UAVi and $\boldsymbol{p_i}$ is the current position of UAVi. The term w_{risk}^i denotes the risk degree of UAV i as perceived within the whole formation. Meanwhile, d_{obs}^i is the smallest distance to the nearest obstacle from the current position, and d_{risk} refers to the risk balance distance parameter. Notably, in situations where a UAV is dangerously close to an obstacle (when $d_{obs}^i \le d_{risk}$), the UAV's target should aim to navigate it away from the obstacle, rather than towards it. Hence, in such cases, we set the risk degree to zero.

5) Continuity Cost: To incorporate the continuity of the formation sequence, the proximate two formation configurations should feature comparable c and s values. Given that the propagation distance remains nearly constant between two formation centers, we place particular emphasis on the angle between them. Each formation configuration in the entire sequence carries a continuity cost c_{con}, which is derived from both the scale continuity and the angle continuity costs:

$$c_{con} = k_{sc} \left| s_{cur} - s_{las} \right| + k_{ac} \left| \theta_{cur} - \theta_{las} \right| \tag{11}$$

where k_{sc} and k_{ac} represent the scale and angle continuity weight parameters respectively, s_{las} denotes the scale of f_c for the preceding formation. The current angle difference θ_{cur}, can be calculated using the direction vector of the current $\boldsymbol{c}$ and the preceding $\mathbf{c}$ of f_c. Similarly, the preceding angle difference, θ_{last}, can be calculated in the same manner.

6) Costs Summary: Thus far, we have discussed a number of evaluation cost functions in (7)-(10), which are pertinent to optimization in the current step, as well as (11), which strives for enhanced performance throughout the entire sequence. The total cost of a single $\boldsymbol{f_c}$ can be calculated as the sum of the above costs.

Remark 1: It is crucial to note that if the current sequence propagation is deemed unsafe, it is terminated to avoid potential conflict with obstacles. Nevertheless, guidance spanning

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a short distance is not our intended aim, as it could lead the entire formation to get trapped in a local minimum. Bearing this in mind, a substantial penalty c_e is imposed on any halted formation sequence before the pre-defined total propagation steps, according to the remaining step number, which is a cost from a holistic perspective of the formation configuration sequence.

C. Task Assignment

The formation targets calculated from $\boldsymbol{f_c}$ could lead to potential collision among UAVs, which could be addressed by the task assignment. Despite considering temporary constant target assignment during the evaluation process, the optimal allocation cannot be guaranteed using such a method, and some intersecting paths may arise from their current positions. Consequently, we need to manage the optimal task assignment within the formation configuration to deconflict the formation. The task assignment problem is formulated as follows:

$$\min_{\sigma} \sum_{i=0}^{N} \left\| \boldsymbol{p}_t^i - \boldsymbol{p}_f^{\sigma(i)} \right\| \tag{12}$$

where σ is the task assignment result (eg.$\sigma(i) = j$).

Problem (12) is defined as a standard linear sum assignment problem, which can be addressed using various prevalent methods, such as the Hungarian algorithm [34], simplex algorithm [35], auction algorithm [36]. In this paper, the auction algorithm is adopted considering the computational load and a desire for optimal results.

D. Algorithm Summary

Up to this point, we have presented all the modules comprising this algorithm. The complete procedure is depicted in Algorithm 1. The algorithm takes several inputs, including the current positions of all UAVs $\mathbf{P}cur$, the composite formation polytope $\mathcal{P}_{all} = \cup \mathcal{P}_i, i \in \{1, \cdots, N\}$, the previous optimal task assignment result σ_0, the initial sampling center $\mathbf{c_0}$, the desired target formation configuration f_c^g, the total number of sampling sequences T_F, and the propagation step T_S. The output of the algorithm is the generated formation path $\Pi_c = \{\pi_1, \ldots, \pi_N\}$ for all UAVs. Employing the leader-follower concept, the position of the leader is assigned as $\mathbf{c_0}$, and the s value in the first member of the previously generated formation configuration sequence is designated as s_0 respectively. The entire calculation process is centralized and executed on the leader UAV. Initially, T_F sampled formation configuration sequences are obtained with multiple iterations of SampleFC(), which is elaborated in Section IV-A. The sequence sampling process is illustrated in Fig. 5. During the sampling process, we assess the cost of each formation configuration and get the current best step with lowest single step cost. The propagation of the current sequence is executed only when the current $\mathbf{f}_{\mathbf{c_i}}^{\mathbf{j}}$ is deemed safe (Lines 4-15). Subsequently, the suspension cost is incorporated and the total cost for each sequence is calculated (Lines 16-17). Besides, the formation configuration sequence with the lowest cost is selected as the result sequence. By solving

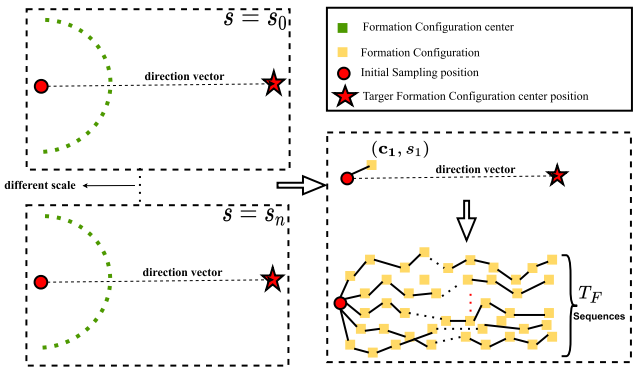


Fig. 5. Sampling process illustration in 2-D space.

Algorithm 1 Sampling-Based Formation Configuration Path Generation

Input: $\mathbf{c_0}$, s_0, $\mathcal{P}_{all}$, σ_0, $\mathbf{P}_{cur}$, $\mathbf{f_c^g}$, T_F, T_S
Output: Π_c
1: **Initialize**($\mathbf{c_0}$,s_0,$\mathcal{P}_{all}$,σ_0,$\mathbf{P}^{cur}$,$\mathbf{f_c^g}$)
2: **for** $i = 0$ to $T_F - 1$ **do**
3: $cost_i \leftarrow 0$, $t_S \leftarrow 0$
4: **for** $j = 0$ to $T_S - 1$ **do**
5: $\mathbf{f_c^j} \leftarrow$ **SampleFC**($\mathbf{c_0}$)
6: $cost_i^j \leftarrow$ **FC_Evaluation**($\mathbf{f_{c_i}^j}$)
7: **if SafeCheck**($\mathbf{f_{c_i}^j}$) **then**
8: $\mathbf{f_{c_i}^j} \leftarrow \mathbf{f_c^j}$
9: $cost_i = cost_i + cost_i^j$
10: $\mathbf{f_c^{last}} \leftarrow \mathbf{f_c^j}$
11: t_S++
12: **else**
13: break
14: **end if**
15: **end for**
16: $cost_i = cost_i +$ **SuspendendPunish**(t_S)
17: $F_{c_i} \leftarrow \{\{f_{c_i}^0, \cdots, f_{c_i}^{p_{step}-1}\}, cost_i\}$
18: **end for**
19: $\mathbf{F_c} \leftarrow$ **SequenceGeneration**($F_{c_0}, \cdots, F_{c_{T_F-1}}$)
20: $\sigma^* \leftarrow$ **TaskAssignment**($\mathbf{F_c}$, $\mathbf{P}_{cur}$)
21: $\Pi_c \leftarrow$ **PathGeneration**($\mathbf{F_c}$, σ^*)
22: **return** Π_c

the task assignment problem, we can obtain the optimal task assignment result based on the first $\mathbf{f_c}$ in $\mathbf{F_c}$ and $\mathbf{P}_{cur}$. Finally, the discrete formation configurations are linked and the result Π_c is obtained with σ^* and $\mathbf{F_c}$ for all UAVs (Line 19-21).

V. Distributed Formation Trajectory Optimization Based on MPPI

The formation path generated in Section IV provides effective guidance for maintaining the formation and avoiding obstacles. However, these paths, denoted as Π_c, consist of discrete waypoints that are connected by straight lines. Consequently, these paths may be too coarse to be directly executed in a real system. In this section, some sub-evaluation objectives of a trajectory are formulated as the components of the cost functions in (5). Additionally, a brief introduction to the Model Predictive Path Integral (MPPI) technique is presented. Finally, the complete trajectory optimization algorithm is summarize.

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A. Objective Functions

For the sake of simplicity, the terminal cost ϕ is set as zero. Besides, the main components of the running cost $\rho(\boldsymbol{x}_t, t)$ presented in (5) are formulated as following:

1) Formation Guidance Cost: To keep the desired formation, the formation guidance cost is introduced to drive the predicted trajectory to be close to the formation path. For UAVi, it is formulated as follow:

$$h_f^i = \sum_{k=0}^{P-1} k_f \left\| \boldsymbol{p}_{\boldsymbol{k}}^{\boldsymbol{i}} - \boldsymbol{p}_{\boldsymbol{d}}^{\boldsymbol{i,k}} \right\| \tag{13}$$

where $k_f \in \mathbb{R}^+$ is the positive weight and $P = T/\Delta t \in \mathbb{R}^+$ is the total propagation step number in the trajectory generation process. For better tracking performance, the T_S in Algorithm 1 is set be equal to P. $\boldsymbol{p}_{\boldsymbol{k}}^{\boldsymbol{i}}$ is the position at step k along the predicted trajectory and $\boldsymbol{p}_{\boldsymbol{d}}^{\boldsymbol{i,k}}$ is the k_{th} formation guidance waypoint extracted from the received guidance path of UAVi.

2) Dynamic Cost: The trajectory is limited by the dynamics of UAV in terms of the maximum velocity and acceleration. For UAVi it is defined as:

$$h_{\mathrm{d}}^i = \begin{cases} k_{\mathrm{dyn}}, & d_{dyn}^i \geq 0 \\ 0, & d_{dyn}^i < 0 \end{cases}$$
$$d_{dyn}^i = sign(\left|v_t^i\right| - v_{max}) + sign(\left|a_t^i\right| - a_{max}) \tag{14}$$

where v_t^i and a_t^i are the velocity and acceleration at time t along the trajectory, v_{max} and a_{max} are the corresponding maximum and $k_{\mathrm{dyn}} \in \mathbb{R}^+$.

3) Smoothness Cost: The smooth trajectory is required in the flight, therefore the smoothness cost is designed to penalize the integral square derivatives of control input $\boldsymbol{u}_t^i$. It is designed as follows:

$$h_S = k_{smo} \int_t^{t+T} {\boldsymbol{u}_\tau^i}^{\mathrm{T}} \boldsymbol{u}_\tau^i d\tau \tag{15}$$

where $k_{smo} \in \mathbb{R}^+$ is the smoothness weight.

4) Obstacle Avoidance Cost: To ensure safety, obstacle avoidance cost is incorporated to keep UAV far away from the obstacles. For UAVi, it is expressed as:

$$h_{safe}^i = \begin{cases} k_{obs}, & d_{obs}^i < d_{obs}^{min} \\ k_{obs}\left(\frac{d_{obs}^{max} - d_{obs}^i}{d_{obs}^{max} - d_{obs}^{min}}\right)^\beta, & d_{obs}^{min} < d_{obs}^i \leq d_{obs}^{max} \\ 0, & d_{obs}^i > d_{obs}^{max} \end{cases} \tag{16}$$

where d_{obs}^{min} and d_{obs}^{max} are the safe threshold of obstacles avoidance, $k_{obs}, \beta > 0 \in \mathbb{R}^+$.

5) Mutual Collision Cost: It is essential to avoid the inner collion during the formation flight. The mutual collision cost is designed to exert a repulsive force between two UAVs within a mutual distance less than d_{mut}^{max}. For UAV i and $j, j \neq i$, it is presented by:

$$h_{mut}^i = \begin{cases} k_{mut}, & d_{mut}^{ij} < d_{mut}^{min} \\ k_{mut}\left(\frac{d_{mut}^{max} - d_{mut}^{ij}}{d_{mut}^{max} - d_{mut}^{min}}\right)^\alpha, & d_{mut}^{min} < d_{mut}^{ij} \leq d_{mut}^{max} \\ 0, & d_{mut}^{ij} > d_{mut}^{max} \end{cases}$$
$$d_{mut}^{ij} = \left\| \Gamma[p_i(t) - p_j(t)] \right\|_2, p_i(t) \in \Omega_i, p_j(t) \in \kappa_j \tag{17}$$

where $k_{mut}, \alpha > 0$, $p_i(t)$ and $p_j(t)$ are the position of UAV i, j at time t from the predicted trajectory Ω_i and the executed trajectory κ_j respectively.Given the downwash peril, a transformation matrix $\Gamma = \mathrm{diag}(1, 1, \lambda), \lambda < 1$ is introduced to transform the euclidean distance to ellipsoidal distance between UAV i and j at time t.

B. Model Predictive Path Integral Control

To make the paper self-contained, in this part, we will give a brief introduction of MPPI, more details can be found in our previous work [37]. The stochastic control problem presented in (6) can be solved with the Hamilton-Jacobo-Bellman(HJB) equation. Thus the HJB equation based on the dynamics model(4) and cost functions(5) is defined as follows:

$$-\frac{\partial V_i}{\partial t} = \rho(\boldsymbol{x}_t^i, t) + \left(A\boldsymbol{x}_t^i\right)^T \frac{\partial V_i}{\partial \boldsymbol{x}_t^i} - \frac{1}{2}\left(\frac{\partial V_i}{\partial \boldsymbol{x}_t^i}\right)^T B R^{-1} B^T \frac{\partial V_i}{\partial \boldsymbol{x}_t^i} + \frac{1}{2}\mathrm{tr}\left(B B^T \frac{\partial^2 V}{\partial {\boldsymbol{x}_t^i}^2}\right) \tag{18}$$

with the boundary condition $V_i(\boldsymbol{x}_{t+T}^i, t+T) = \phi(\boldsymbol{x}_{t+T}, t+T) = 0$. The optimal control input is expressed as follows:

$$\boldsymbol{u}_t^i = -R^{-1} B^T \frac{\partial V_i}{\partial \boldsymbol{x}_t^i} \tag{19}$$

By solving the partial differential equation (PDE) of cost function $V_i(\boldsymbol{x}_t^i, t)$ to state vector $\boldsymbol{x}_t^i$, $\boldsymbol{u}_t^i$ is obtained. However, it is nontrivial to solve such a problem with high dimension of $\boldsymbol{x}_t^i$ with tradition methods. Inspired by the path integral control framework, the PDE can be transformed into a path integral. For the sake of utilization of the Fetnman-Kac formula relating PDE to path integrals, (18) is needed to by linear. Thus (18) is linearized with exponential transform:

$$\frac{\partial \Psi_i}{\partial t} = \frac{\Psi_i}{\lambda}\rho(\boldsymbol{x}_t^i, t) - \left(A\boldsymbol{x}_t^i\right)^T \frac{\partial \Psi_i}{\partial \boldsymbol{x}_t^i} - \frac{1}{2} tr\left(B B^T \frac{\partial^2 \Psi_i}{\partial {\boldsymbol{x}_t^i}^2}\right)$$
$$\Psi_i\left(\boldsymbol{x}_t^i, t\right) = e^{-\frac{V_i(\boldsymbol{x}_t^i, t)}{\lambda}} \tag{20}$$

where $\lambda \in \mathbb{R}^+$. Then the Fetnman-Kac formula is applied to (18) denoted by:

$$\Psi_i\left(\boldsymbol{x}_t^i, t\right) = \mathbb{E}[exp(-\frac{1}{\lambda})]S(t)$$
$$S(t) = \int_t^{t+T} \rho(\boldsymbol{x}_{\boldsymbol{\tau}}^{\boldsymbol{i}}, \tau) dt \tag{21}$$

Now the path integral form for the cost function in (21) is obtained through the above process. By computing the gradient of $\Psi_i\left(\boldsymbol{x}_t^i, t\right)$ with respect to the initial state $\boldsymbol{x}_t^i$ analytically, we can get follows:

$$\boldsymbol{u}_t^{i*} dt = R^{-1} B^T \left(B R^{-1} B^T\right)^{-1} \frac{\mathbb{E}\left[\exp\left(-\frac{S(t)B}{\lambda} d\boldsymbol{v}_t^i\right)\right]}{\mathbb{E}\left[\exp\left(-\frac{S(t)}{\lambda}\right)\right]} \tag{22}$$

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Subsequently, the dynamic model for trajectory sampling is discretized as follows:

$$d\boldsymbol{x_{t+1}} = A\boldsymbol{x_t}\Delta t + B\mu_{\mathbf{t}}\Delta t \tag{23}$$

Finally, the discretized expression of $\boldsymbol{u}_t^{i*}$ is derived:

$$\boldsymbol{u}_t^{i*} = \boldsymbol{u}_{t-1}^{i} + \frac{\mathbb{E}\left[\exp\left(-\frac{S(t+T)}{\lambda}d\boldsymbol{v}_t^i\right)\right]}{\mathbb{E}\left[\exp\left(-\frac{S(t+T)}{\lambda}\right)\right]} \tag{24}$$

where $S(t+T) = \sum_{m=0}^{R}\rho(\boldsymbol{x}_{m,t}^i, t)\Delta t$, Δt is the time interval and R is the total number of sampled trajectories.

C. Distributed Formation Trajectory Optimization

As delineated through the aforementioned predictive inference control process, it is feasible to ascertain the optimal control input. Consequently, this leads to the extraction of the optimal trajectory, which is fundamentally based on the state vector $\boldsymbol{x_t}$ at the time t and the optimal control sequence U_i^*.

For the UAVi, the trajectory optimization approach, as proposed in this study, is succinctly summarized in Algorithm 2. The algorithm comprises several variables: the current state vector $\boldsymbol{x}_t^i$, initial control input $\boldsymbol{u}_{init}^i$, initial control input sequence U_i^{init}, total propagation steps P, R and the alternative executed trajectories set signified as $\mathbb{K} = \{\kappa_j, \ldots, \kappa_N\}, j \neq i$. The output of the algorithm is the optimally generated trajectory κ_i by UAVi. The algorithm initiates by setting the initial state vector $\boldsymbol{x}_0^i$ equal to $\boldsymbol{x}_t^i$. R distinct trajectories are generated by employing the random sampling set $\mathbb{V}^i$ where the mth trajectory is explicitly denoted as $\mathbb{V}_m^i$. These trajectories are generated parallelly over P steps integrating both the control input $\boldsymbol{u}_t^i$ and disturbance $\boldsymbol{v}_t^i$. Simultaneously, the trajectories undergo assessment in the mth Monte Carlo sampling process by the function $S(\mathbb{V}_m^i)$ (Line 2-8).

Subsequently, the importance weight $\omega(\mathbb{V}_m^i)$ is computed for each trajectory based on the expectation cost (Line 9-13), leading to the acquisition of optimal control inputs (Line 9-17). The states of the optimal trajectory are then obtained by incorporating u^* within the dynamic model. Moreover, the process involves the iterative update of the optimal control sequence, where the current U_i^* is employed as the initial control sequence in the succeeding calculation round (Line 20-24). To amplify the practicality of the method, the obtained trajectory states undergo the processes of connection and smoothing operations, thereby yielding the final optimal trajectory as the output (Line 25).

Algorithm 2 Distributed Multi-UAV Trajectory Optimization Based on MPPI

Input: $\boldsymbol{x}_t^i$,$\boldsymbol{u}_{init}^i$, U_i^{init}, P, R, $\mathbb{K}$
Output: κ_i
1: $x_0^i = \boldsymbol{x}_t^i$
2: **for** $m = 0$ to $R-1$ **do**
3: $\quad\mathbb{V}_m^i \leftarrow \{\boldsymbol{v}_{\boldsymbol{m,0}}^{\boldsymbol{i}}, \boldsymbol{v}_{\boldsymbol{m,1}}^{\boldsymbol{i}}, \ldots, \boldsymbol{v}_{\boldsymbol{m,P-1}}^{\boldsymbol{i}}\}$
4: $\quad$**for** $n = 0$ to $P-1$ **do**
5: $\quad\quad\mathbf{x}_{\mathbf{n+1}}^{\mathbf{i}} = A\boldsymbol{x}_{\boldsymbol{n}}^{\boldsymbol{i}} + B(\boldsymbol{u}_t^i + \boldsymbol{v}_t^{\boldsymbol{i}})$
6: $\quad\quad S(\mathbb{V}_m^i) += S(\mathbb{V}_m^i) + \rho(\boldsymbol{x}_{\boldsymbol{n+1}}^{\boldsymbol{i}}, t_n)$
7: $\quad$**end for**
8: **end for**
9: $S_{min} \leftarrow \min\{S(\mathbb{V}_0^i), S(\mathbb{V}_1^i), \ldots S(\mathbb{V}_{R-1}^i)\}$
10: $\varepsilon \leftarrow \sum_{m=0}^{R} exp(-\frac{1}{\lambda}(S(\mathbb{V}_m^i) - S_{min}))$
11: **for** $m = 0$ to $R-1$ **do**
12: $\quad\omega(\mathbb{V}_m^i) \leftarrow \frac{1}{\varepsilon}exp(-\frac{1}{\lambda}(S(\mathbb{V}_m^i) - S_{min}))$
13: **end for**
14: **for** $n = 0$ to $P-1$ **do**
15: $\quad\boldsymbol{u}_{\boldsymbol{n}}^{\boldsymbol{i}}{}^* = \boldsymbol{u}_{\boldsymbol{n}}^{\boldsymbol{i}} + \sum_{m=0}^{R-1} w(\mathbb{V}_m^i)\boldsymbol{v}_{\boldsymbol{m,n}}^{\boldsymbol{i}}$
16: **end for**
17: **for** $n = 0$ to $P-1$ **do**
18: $\quad\boldsymbol{x}_{\boldsymbol{n+1}}^{\boldsymbol{i}}{}^* = A\boldsymbol{x}_{\boldsymbol{n}}^{\boldsymbol{i}}{}^* + B\boldsymbol{u}_t^{i*}$
19: **end for**
20: **for** $n = 0$ to $P-2$ **do**
21: $\quad\boldsymbol{u}_{\boldsymbol{n}}^{\boldsymbol{i}}{}^* = \boldsymbol{u}_{\boldsymbol{n+1}}^{\boldsymbol{i}}{}^*$
22: **end for**
23: $\boldsymbol{u}_{\boldsymbol{P-1}}^{\boldsymbol{i}}{}^* = \boldsymbol{u}_{init}^i$
24: $U_i^* = \{\boldsymbol{u}_{\mathbf{0}}^{\boldsymbol{i}}{}^*, \ldots, \boldsymbol{u}_{\boldsymbol{P-1}}^{\boldsymbol{i}}{}^*\}$
25: return $\kappa_i \leftarrow$ **Link**$(\boldsymbol{x}_{\mathbf{1}}^{\boldsymbol{i}}{}^*, \boldsymbol{x}_{\mathbf{2}}^{\boldsymbol{i}}{}^*, \ldots, \boldsymbol{x}_{\boldsymbol{P}}^{\boldsymbol{i}}{}^*)$

VI. Result Analysis

In this section, the proposed strategy is validated in some simulation scenarios. Firstly, the criteria for assessing the safety and formation-keeping performance of the formation flight are presented. The relationship between formation safety and formation keeping is then discussed in depth to clarify the functionality of the proposed method. Subsequently, comparative simulations are conducted between the proposed method and the approaches presented in [20] and [22]. Finally, an analysis of the computational time and bandwidth on onboard devices is provided, demonstrating the potential of the proposed method in real-world applications.

A. Evaluation Criteria

The goal of the proposed algorithm is to keep the whole formation safe and fly in the desired shape. Therefore, the formation performance is evaluated from the perspective of safety and formation similarity. More details of formation similarity can be seen in [22]. To evaluate the whole process of formation flight, the average formation similarity error $\overline{e_f}$ and the max formation similarity error e_f^{max} which describes the comprehensive formation keeping and the degree of formation distortion are defined as follows:

$$\begin{aligned}\overline{e_f} &= \frac{\int_{t_0}^{t_e} e_f(t)dt}{t_e - t_0}\\ e_f^{max} &= max e_f(t), t \in [t_0, t_{end}]\end{aligned} \tag{25}$$

where the definition of formation similarity error e_f can be found in [22], $e_f(t)$ is the formation similarity error at time t, and t_0, t_e are the formation start time and end time during the whole flight process.

Additionally, to assess the safety of the formation, a parameter termed the formation minimum safe distance d_{obs}^f is introduced. This metric represents the smallest real-time distance between any UAV in the formation and the nearest obstacle.

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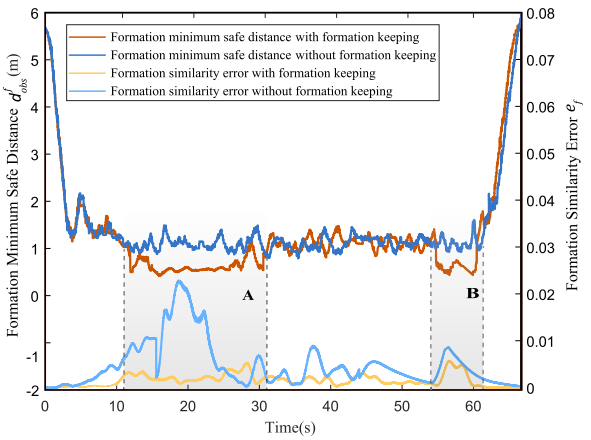


Fig. 6. Comparison of formation flight in a clustered environment with and with formation keeping.

B. The Relationship Between the Safety and Formation Keeping

To investigate the relationship between formation safety and formation keeping, as well as to evaluate the effectiveness of the proposed approach, a comparative simulation experiment is conducted in a clustered environment with six UAVs. The experiment compares the performance of methods with and without formation-keeping strategy. The method with formation keeping is the proposed approach in this work. To isolate the influence of formation keeping, the front-end algorithm is excluded, and the same back-end algorithm described in Section V is employed. The safety of the formation and the effectiveness of formation keeping are evaluated in real time using d_{obs}^f and e_f with the results presented in Fig.6. Throughout most time of the simulation, formation keeping and safety do not interfere with each other, and the safety levels remain comparable between the methods with and without formation keeping. However, during periods of conflict between obstacle avoidance and formation keeping, which are highlighted in time region A (left gray region) and time region B (right gray region), the formation keeping strategy may slightly reduce the safety margin to prevent a significant increase in the formation similarity error. This adjustment helps maintain the integrity of the formation shape. While this approach reduces the flexibility of individual UAVs and slightly lowers d_{obs}^f, all UAVs maintain a sufficient d_{obs}^f to ensure safe navigation and preservation of the desired formation shape.

C. Comparison Experiments With Different Methods

To evaluate the performance of the proposed method, comparison experiments are conducted with two state-of-the-art methods [20], [22]. Key metrics, including the minimum d_{obs}^f during the whole flight (d_{safe}), the average formation similarity error ($\overline{e_f}$), the maximum formation similarity error (e_f^{max}), the computation time for the optimal trajectory (t_{sol}) and the total flight time (t_{fly}) are calculated for quantitative comparisons. The maximum flight velocity is set to 1.5 m/s and all simulations are performed on a computer equipped with an Intel i7-12700F CPU and NVIDIA GeForce RTX 3070 GPU. Additionally, certain parameters in Algorithm 2 are tuned,

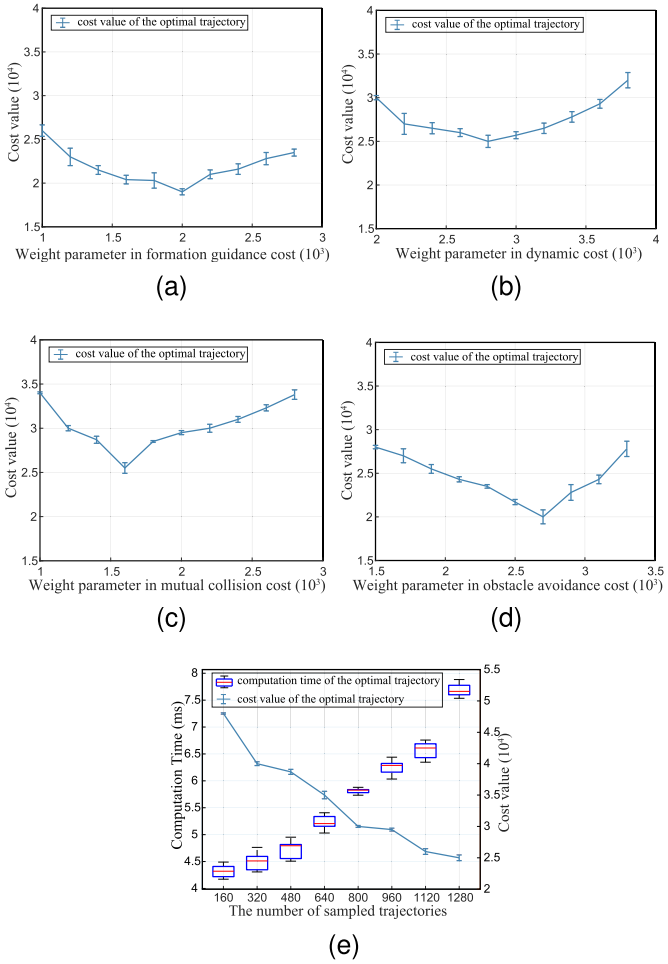


Fig. 7. The main parameters tuning process of model predictive path integral based distributed trajectory optimization. Fig. 7a-Fig.7d are the tuning process of the weight parameters in some cost functions. The cost of the optimal trajectory serves as a performance metric for the generated trajectories, enabling the evaluation of parameter influence on the algorithm. Initially, k_{smo} is set as 1000. Subsequently, the parameters k_{f}, k_{dyn}, k_{mut} and k_{obs} are tuned iteratively using a controlled variable method. Finally, the optimal weight parameter set is tested with varying values of R to determine the best value of R, as demonstrated in Fig. 7e.

as illustrated in Fig. 7. By balancing the trade-off between computational cost and optimization performance, the specific parameters used in the proposed algorithm are determined and the values of them are detailed in Table I.

The desired formation for this study is a triangular configuration involving six UAVs. Simulations are conducted across a variety of environments, including clustered scenarios with dimensions of 50×40 m, containing varying densities of pillar obstacles categorized as Sparse (50 pillars), Medium (100 pillars), and Dense (150 pillars). Additionally, simulations are performed in a narrow corridor environment measuring 26×8 m to evaluate the scalability and adaptability of the proposed formation maintenance methods. These are collectively referred to as different-environment experiments. Beyond the experiments involving a fixed six-UAV triangular formation, additional triangular configurations with 8, 10, and 13 UAVs are also examined to assess the effectiveness of the proposed approach under different formation sizes, which is noted as different-formation experiment.

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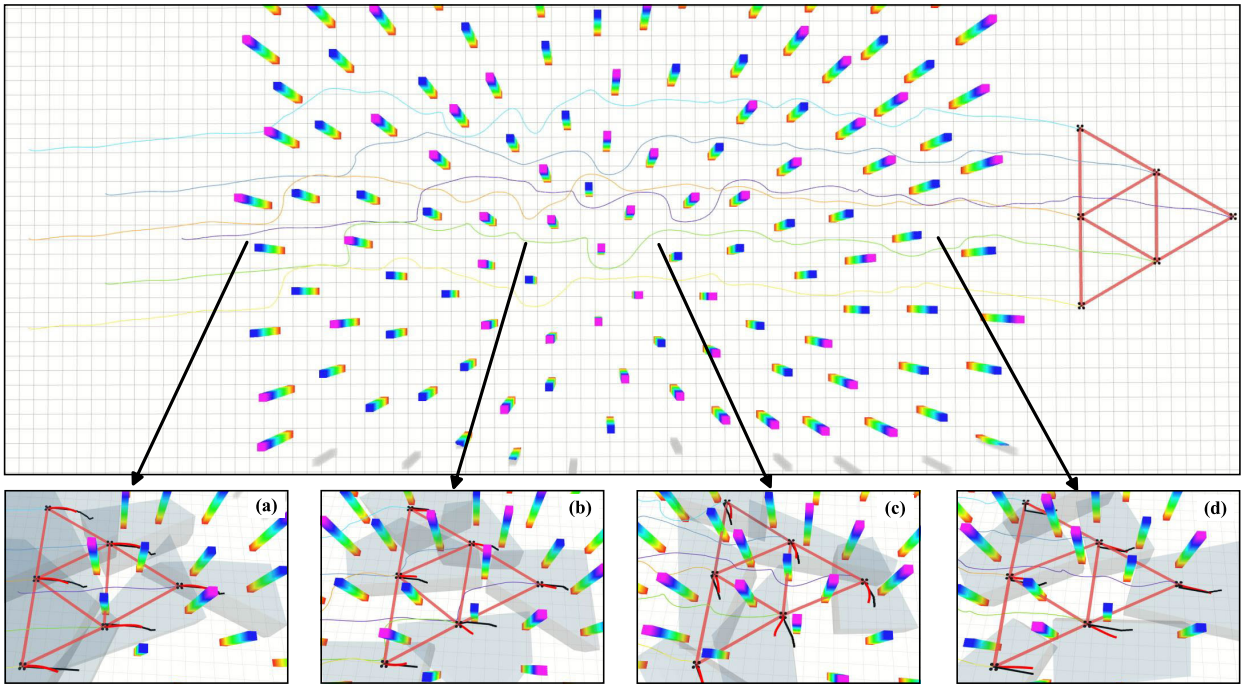


Fig. 8. A triangular formation, consisting of six uncrewed aerial vehicles (UAVs), traverses through an unknown environment with obstacles from the left side to the right side. (a)-(d) are the snapshots in the fixed time interval.

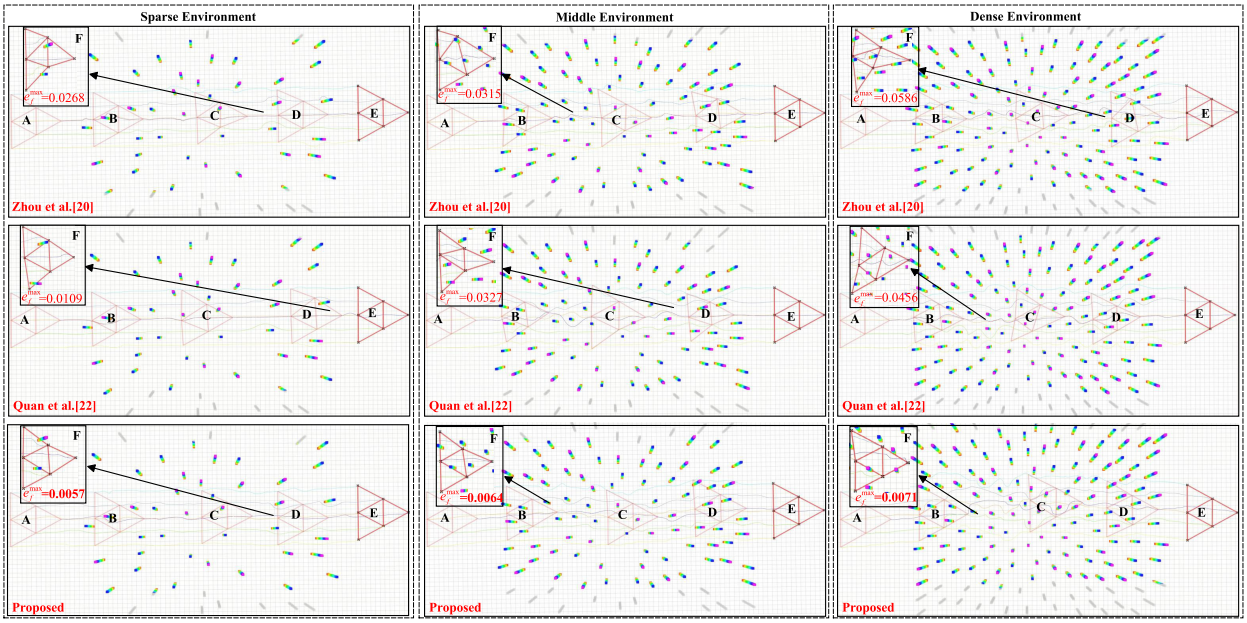


Fig. 9. The formation flight process generated by our proposed method, [20] and [22] in environments with different number of obstacles. The shallow red triangle A represents the initial formation in the process, while B, C, and D correspond to the formation traces on the left, middle, and right sides of the obstacle scenarios, respectively. The red triangle E denotes the final state of the formation. The positions with e_f^{max} are indicated by arrows, with each corresponding formation at these positions shown as red triangle F. The formation with the proposed method shows the lowest e_f^{max} at all scenarios.

As illustrated in Fig. 8, the formation employing our proposed method successfully navigates through an unknown environment containing obstacles. The comparison processes of the simulations in the cluster environments and the narrow corridor are depicted in Fig. 9 and Fig. 10. The flight results of different-environment experiment and different-formation experiment are summarized in Table II and Table III respectively. It can be observed that formations with all methods, including the proposed method and the ones presented in [20] and [22], successfully navigate through the clustered environments while maintaining the formation. With the adjustment ability of the formation keeping in response to obstacles that distort its shape, d_{safe} with the proposed method is slightly lower compared to the other two methods. However, it remains sufficient for effective obstacle avoidance. The formation keeping is presented as a formation vector field in [20] and one aspect of gradient optimization problem in [22] respectively, which is optimized concurrently with other aspects,

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TABLE I
PARAMETER LIST IN ALGORITHM 2

Parameter	Value
The number of sampled trajectories R	1120
The propagation step of a sampled sequence P	35
Weight parameter in smoothness cost k_{smo}	1000
Weight parameter in formation guidance cost k_{f}	2000
Weight parameter in dynamic cost k_{dyn}	2700
Distance parameter in mutual collision cost d_{mut}^{min}	0.5
Distance parameter in mutual collision cost d_{mut}^{max}	1.2
Weight parameter in mutual collision cost k_{mut}	1600
Exponential coefficient in mutual collision cost α	1.5
Distance parameter in obstacle avoidance cost d_{obs}^{max}	0.8
Distance parameter in obstacle avoidance cost d_{obs}^{min}	0.4
Weight parameter in obstacle avoidance cost k_{obs}	2600
Exponential coefficient in obstacle avoidance cost β	1.8

TABLE II
THE PERFORMANCE COMPARISON IN DIFFERENT ENVIRONMENTS

Env.	Method	$d_{safe}(m)$	$\overline{e_f}$	e_f^{max}	$t_{sol}(ms)$	$t_{fly}(s)$
Sparse	[20]	0.86	0.0025	0.0268	**5.3**	69.3
	[22]	0.95	0.0013	0.0109	6.3	67.8
	Proposed	**1.1**	**0.0011**	**0.0057**	7.2	**66.6**
Middle	[20]	0.81	0.0036	0.0315	**5.8**	75.8
	[22]	**0.83**	0.0052	0.0327	6.7	78.2
	Proposed	0.68	**0.0012**	**0.0064**	7.4	**74.9**
Dense	[20]	0.74	0.0059	0.0586	**6.4**	82.7
	[22]	**0.76**	0.0094	0.0456	7.4	79.3
	Proposed	0.62	**0.0020**	**0.0071**	7.9	**78.2**
Corridor	[20]	0	-	-	-	-
	[22]	0.59	0.0102	0.0492	**5.9**	46.3
	Proposed	**0.72**	**0.0007**	**0.0134**	6.5	**44.5**

TABLE III
THE PERFORMANCE COMPARISON WITH DIFFERENT NUMBERS OF UAV

UAV	Method	$d_{safe}(m)$	$\overline{e_f}$	e_f^{max}	$t_{sol}(ms)$	$t_{fly}(s)$
N=8	[20]	0.96	0.0038	0.0275	**6.3**	75.6
	[22]	0.98	0.0024	0.0135	6.8	76.2
	Proposed	**1.15**	**0.0013**	**0.0089**	8.3	**74.8**
N=10	[20]	0.63	0.0043	0.0337	**6.7**	74.6
	[22]	0.62	0.0046	0.0302	7.3	75.6
	Proposed	**0.65**	**0.0015**	**0.0093**	8.8	**76.1**
N=13	[20]	0.66	0.0059	0.0579	**7.3**	79.5
	[22]	**0.69**	0.0683	0.0456	7.8	79.4
	Proposed	0.58	**0.0016**	**0.0131**	9.4	**77.1**

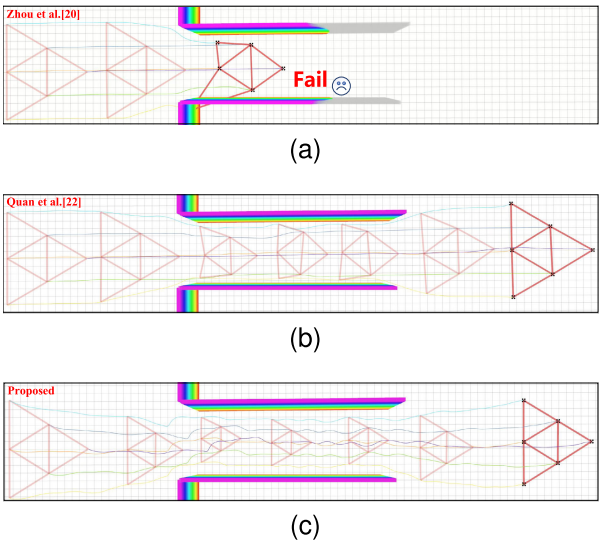


Fig. 10. The shallow red line are the formation trace left every fixed distance. Fig 10a, Fig 10b and Fig 10c are the the flight process with the method in [20] and [22] and the proposed method respectively.

such as collision evasion, obstacles avoidance and trajectory smoothness. Consequently, the influence of formation constraints can be diminished to prioritize individual security. The performance of formation maintenance deteriorates as the number of obstacles increases in the methods presented by [20] and [22], as evidenced by the significant increase of $\overline{e_f}$ and e_f^{max} in different clustered scenarios, which is displayed in Table II. In the proposed method, the issue of weakened formation preservation is circumvented, as the sampled formation configurations are confined to the local safe region, thereby ensuring formation connectivity and obstacle avoidance as well. As shown in Fig. 9, the formation obtained using the proposed method exhibits minimal deformation across various environments compared to the other two methods.

Moreover, the narrow corridor presents a significant challenge for the methods, as it requires the adjustment of the formation scale. The method in [20] lacks the ability to adjust the scale, as it constructs vector fields with a constant formation size. This limitation prevents adaptation to a smaller formation that maintains a similar shape, ultimately leading to the failure event as shown in Fig. 10a. The method in [22] can adjust the scale with a limitation, but the obstacles avoidance weakens the formation-keeping performance in the corridor, resulting in the near-distorted formation shape in this scenario. In contrast, when the safe region within the confined corridor is too restricted, the scale of the formation is adeptly adjusted to as low as 0.5, thereby exemplifying the superior adaptability of the formation in challenging environments, as achieved by the proposed method. Although the computational efficiency is improved by the co-optimization method in [20] and [22], the formation fails to consistently maintain the desired shape across all experiments. With the effectiveness of the formation guidance path, $\overline{e_f}$ and e_f^{max} are significantly reduced compared to the methods in [20] and [22] in both the different-environment experiments and the different-formation experiments. Furthermore, the flight time t_{fly} is slightly decreased with the proposed method. In conclusion, the proposed method demonstrates superior performance in environments with varying obstacles as well as in scenarios involving different desired formations.

D. Potential for Real-World Application

In this part, the feasibility of the proposed algorithm on the real-world scenario is discussed. Unlike typical consumer-grade computers used in everyday applications, the onboard computer utilized on real UAVs has limited computational resources. Additionally, communication between these UAVs relies on onboard communication devices, which are also subject to bandwidth limitations. To evaluate the potential of deploying the proposed algorithm in real-world scenarios,

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TABLE IV
THE COMPUTATIONAL TIME AND RESOURCE USAGE ON DIFFERENT DEVICES

Device	$t_{front}(ms)$	$t_{back}(ms)$	$O_{CPU}(\%)$	$O_{GPU}(\%)$
Xavier NX	6.9	13.8	32.4	37.6
Orin NX	5.2	11.6	30.2	35.7
AGX Orin	3.6	8.4	25.9	30.4

TABLE V
THE BANDWIDTH OF SOME SHARED MESSAGES

Environment	B_κ	$B_\mathcal{P}$	$B\pi$
Sparse	10.5 KB/s	11.3 KB/s	9.4 KB/s
Middle	10.3 KB/s	11.8 KB/s	9.2 KB/s
Dense	10.8 KB/s	12.4 KB/s	9.5 KB/s
Corridor	10.6 KB/s	10.8 KB/s	9.6 KB/s

several tests are conducted on commonly used onboard computers, including Xavier NX, Orin NX and AGX Orin. The computational time is divided into two part: the consumption time on front-end formation guidance paths generation t_{front} and on the back-end MPPI-based trajectory optimization t_{back}. The resource utilization on CPU and GPU are noted as O_{CPU} and O_{GPU} respectively. Each device operates as a single UAV, while the simulation experiments are provided by a ground station. Three UAVs are utilized collaboratively in these experiments, with communication between them facilitated via onboard communication devices. The computational results are summarized in Table IV. It can be observed that both t_{front} and t_{back} are consistently below 15 ms on all onboard computers, and the resource utilization percentages remain below 40%, thereby ensuring sufficient computational capacity for other modules.

During the process of experiments, the average bandwidths of the essential shared messages are monitored with the ROS tool "rostopic bw". The shared messages include the executable trajectory, the safe flight corridor, and the formation guidance path, whose average bandwidths are noted as B_κ, $B_\mathcal{P}$ and $B\pi$ respectively. The bandwidth results in different scenarios are shown in Table V. The bandwidth of each message is approximately 10 KB/s. Notably, $B_\mathcal{P}$ increases with the number of obstacles, due to the corresponding rise in the number of hyperplanes. Nevertheless, the bandwidth requirements remain feasible for transmission using real-world communication devices. In conclusion, the proposed method is well-suited for real-world applications, providing the benefits of real-time computation and low bandwidth requirements.

VII. CONCLUSION

In this paper, a sampling-based hierarchical trajectory planning method is proposed, consisting of front-end and back-end approaches, to achieve safe UAV formation flight with the desired shape. The formation safe region can be defined by transmitting the safe flight corridor, which represents the limited safe area with a small data volume. Subsequently, the formation guidance paths are derived through a sampling-based front-end path generation method, wherein numerous formation configurations are stochastically sampled. The optimal sequence of formation configurations is then identified through the application of specific cost functions, from which the guidance paths are generated. Furthermore, a distributed trajectory optimization technique based on MPPI is introduced as the back-end method, which guarantees the generation of smooth and safe trajectories. Owing to the parallel computational capabilities afforded by the GPU, the proposed sampling-based method is capable of functioning in real time. Finally, extensive simulations are conducted to validate the proposed framework, and comparisons are made with two other state-of-the-art algorithms. The result demonstrates that the proposed method improves the overall performance of the formation in environments with challenging obstacles. In the future, we aim to further develop the proposed method into a fully distributed approach and apply it to real-world scenarios, such as goods delivery in unknown environments.

REFERENCES

[1] D. Liu, X. Zhu, W. Bao, B. Fei, and J. Wu, "SMART: Vision-based method of cooperative surveillance and tracking by multiple UAVs in the urban environment," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 24941–24956, Dec. 2022.

[2] L. Gonçalves and B. Damas, "Automatic detection of rescue targets in maritime search and rescue missions using UAVs," in *Proc. Int. Conf. Unmanned Aircr. Syst. (ICUAS)*, Jun. 2022, pp. 1638–1643.

[3] K. Dorling, J. Heinrichs, G. G. Messier, and S. Magierowski, "Vehicle routing problems for drone delivery," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 47, no. 1, pp. 70–85, Jan. 2017.

[4] J. Zhao, J. Zhang, D. Li, and D. Wang, "Vision-based anti-UAV detection and tracking," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 12, pp. 25323–25334, Dec. 2022.

[5] Y. Hui, X. Zhang, H. Shen, H. Lu, and B. Tian, "DPPM: Decentralized exploration planning for multi-UAV systems using lightweight information structure," *IEEE Trans. Intell. Vehicles*, vol. 9, no. 1, pp. 613–625, Jan. 2024.

[6] W. Ren and E. Atkins, "Distributed multi-vehicle coordinated control via local information exchange," *Int. J. Robust Nonlinear Control*, vol. 17, nos. 10–11, pp. 1002–1033, Jul. 2007.

[7] X. Wang, B. Zerr, H. Thomas, B. Clement, and Z. Xie, "Pattern formation of multi-AUV systems with the optical sensor based on displacement-based formation control," *Int. J. Syst. Sci.*, vol. 51, no. 2, pp. 348–367, Jan. 2020.

[8] S.-M. Kang, M.-C. Park, B.-H. Lee, and H.-S. Ahn, "Distance-based formation control with a single moving leader," in *Proc. Amer. Control Conf.*, Jun. 2014, pp. 305–310.

[9] Z. Han, L. Wang, Z. Lin, and R. Zheng, "Formation control with size scaling via a complex Laplacian-based approach," *IEEE Trans. Cybern.*, vol. 46, no. 10, pp. 2348–2359, Oct. 2016.

[10] M. Mesbahi, "On a dynamic extension of the theory of graphs," in *Proc. Amer. Control Conf.*, vol. 2, May 2002, pp. 1234–1239.

[11] H. Garcia de Marina, "Distributed formation maneuver control by manipulating the complex Laplacian," *Automatica*, vol. 132, Oct. 2021, Art. no. 109813.

[12] Z. Wang, S. B. Andersson, and R. Tron, "Bearing-based formation control with optimal motion trajectory," in *Proc. Amer. Control Conf. (ACC)*, Jun. 2022, pp. 486–493.

[13] S. Zhao, "Affine formation maneuver control of multiagent systems," *IEEE Trans. Autom. Control*, vol. 63, no. 12, pp. 4140–4155, Dec. 2018.

[14] M. Aranda and A. Perez-Yus, "Affine multirobot formation control based on a modular team structure," *IEEE Robot. Autom. Lett.*, vol. 9, no. 6, pp. 5807–5814, Jun. 2024.

[15] K. Fathian, S. Safaoui, T. H. Summers, and N. R. Gans, "Robust 3D distributed formation control with collision avoidance and application to multirotor aerial vehicles," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2019, pp. 9209–9215.

[16] M. Turpin, N. Michael, and V. Kumar, "Trajectory design and control for aggressive formation flight with quadrotors," *Auto. Robots*, vol. 33, nos. 1–2, pp. 143–156, Aug. 2012.

[17] A. Weinstein, A. Cho, G. Loianno, and V. Kumar, "Visual inertial odometry swarm: An autonomous swarm of vision-based quadrotors," *IEEE Robot. Autom. Lett.*, vol. 3, no. 3, pp. 1801–1807, Jul. 2018.

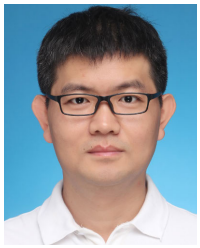
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[18] X. Zhang, F. Zhang, and P. Huang, "Formation planning for tethered multirotor UAV cooperative transportation with unknown payload and cable length," *IEEE Trans. Autom. Sci. Eng.*, vol. 21, no. 3, pp. 3449–3460, Jul. 2023.
[19] Y. Wang, Y. Yue, M. Shan, L. He, and D. Wang, "Formation reconstruction and trajectory replanning for multi-UAV patrol," *IEEE/ASME Trans. Mechatronics*, vol. 26, no. 2, pp. 719–729, Apr. 2021.
[20] D. Zhou, Z. Wang, and M. Schwager, "Agile coordination and assistive collision avoidance for quadrotor swarms using virtual structures," *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 916–923, Aug. 2018.
[21] D. Morgan, G. P. Subramanian, S.-J. Chung, and F. Y. Hadaegh, "Swarm assignment and trajectory optimization using variable-swarm, distributed auction assignment and sequential convex programming," *Int. J. Robot. Res.*, vol. 35, no. 10, pp. 1261–1285, Sep. 2016.
[22] L. Quan, L. Yin, C. Xu, and F. Gao, "Distributed swarm trajectory optimization for formation flight in dense environments," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2022, pp. 4979–4985.
[23] W. Liu, J. Hu, H. Zhang, M. Y. Wang, and Z. Xiong, "A novel graph-based motion planner of multi-mobile robot systems with formation and obstacle constraints," *IEEE Trans. Robot.*, vol. 40, pp. 714–728, 2024.
[24] D. Roy, A. Chowdhury, M. Maitra, and S. Bhattacharya, "Multi-robot virtual structure switching and formation changing strategy in an unknown occluded environment," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2018, pp. 4854–4861.
[25] J. Alonso-Mora, S. Baker, and D. Rus, "Multi-robot navigation in formation via sequential convex programming," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2015, pp. 4634–4641.
[26] J. Alonso-Mora, E. Montijano, M. Schwager, and D. Rus, "Distributed multi-robot formation control among obstacles: A geometric and optimization approach with consensus," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2016, pp. 5356–5363.
[27] J. Alonso-Mora, S. Baker, and D. Rus, "Multi-robot formation control and object transport in dynamic environments via constrained optimization," *Int. J. Robot. Res.*, vol. 36, no. 9, pp. 1000–1021, 2017.
[28] H. Zhu, J. Juhl, L. Ferranti, and J. Alonso-Mora, "Distributed multi-robot formation splitting and merging in dynamic environments," in *Proc. Int. Conf. Robot. Autom. (ICRA)*, May 2019, pp. 9080–9086.
[29] R. A. Newcombe et al., "KinectFusion: Real-time dense surface mapping and tracking," in *Proc. 10th IEEE Int. Symp. Mixed Augmented Reality*, Oct. 2011, pp. 127–136.
[30] H. Oleynikova, Z. Taylor, M. Fehr, R. Siegwart, and J. Nieto, "Voxblox: Incremental 3D Euclidean signed distance fields for on-board MAV planning," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2017, pp. 1366–1373.
[31] B. Zhou, F. Gao, L. Wang, C. Liu, and S. Shen, "Robust and efficient quadrotor trajectory generation for fast autonomous flight," *IEEE Robot. Autom. Lett.*, vol. 4, no. 4, pp. 3529–3536, Oct. 2019.
[32] Q. Wang et al., "Fast iterative region inflation for computing large 2-D/3-D convex regions of obstacle-free space," 2024, *arXiv:2403.02977*.
[33] H. J. Kappen, "An introduction to stochastic control theory, path integrals and reinforcement learning," *AIP Conf. Proc.*, vol. 887, pp. 149–181, Jan. 2007.
[34] S. Chopra, G. Notarstefano, M. Rice, and M. Egerstedt, "A distributed version of the Hungarian method for multirobot assignment," *IEEE Trans. Robot.*, vol. 33, no. 4, pp. 932–947, Aug. 2017.
[35] M. Bürger, G. Notarstefano, F. Bullo, and F. Allgöwer, "A distributed simplex algorithm for degenerate linear programs and multi-agent assignments," *Automatica*, vol. 48, no. 9, pp. 2298–2304, Sep. 2012.
[36] D. P. Bertsekas and D. A. Castanon, "The auction algorithm for the transportation problem," *Ann. Oper. Res.*, vol. 20, no. 1, pp. 67–96, Dec. 1989.
[37] H. Lu, Q. Zong, S. Lai, B. Tian, and L. Xie, "Flight with limited field of view: A parallel and gradient-free strategy for micro aerial vehicle," *IEEE Trans. Ind. Electron.*, vol. 69, no. 9, pp. 9258–9267, Sep. 2022.



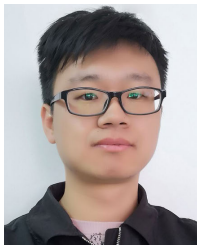
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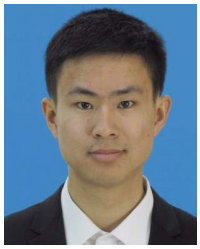
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